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Inventor(s)

KIPFMILLER; Eric

CATALYTIC CONVERTER ANTI-THEFT SYSTEMS AND METHODS

Abstract

A catalytic converter anti-theft system for an exhaust system of a vehicle includes a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system. A cable is configured to be disposed about the thermally insulating first layer. A cut and abrasion resistant second layer is configured to be disposed about the cable. An abrasion resistant third layer is configured to be disposed about the cut and abrasion resistant second layer.

Inventors: KIPFMILLER; Eric (Auburn Hills, MI)

Applicant: Penske Truck Leasing Co., L P. (Reading, PA)

Family ID: 96948954

Assignee: Penske Truck Leasing Co., L P. (Reading, PA)

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Background/Summary

BACKGROUND

[0001] Catalytic converters are commonly detached from vehicles and stolen because they are constructed from valuable metals, such as rhodium, palladium, and platinum, which can be recycled and resold for several to tens of thousands of dollars per ounce. These thefts cost the automotive industry tens of millions of dollars each year.

[0002] Thieves typically detach catalytic converters from vehicles by cutting through exhaust conduits on opposite sides of the catalytic converters. Accordingly, some previous anti-theft systems attempted to deter and inhibit thieves by at least partially surrounding the exhaust conduits and/or catalytic converters with one or more cables. However, such systems typically require significant amounts of time to install. In addition, some saws can nevertheless be maneuvered through spaces between cable sections, and the exhaust conduits may nevertheless be cut, albeit with more difficulty.

[0003] It is with respect to these and other general considerations that embodiments have been described. Also, although relatively specific problems have been discussed, it should be understood that the embodiments should not be limited to solving the specific problems identified in the background.

SUMMARY

[0004] Aspects of the present disclosure are directed to catalytic converter anti-theft systems and methods that deter and/or inhibit cutting adjacent exhaust conduits (for example, using powered saws, grinders, snips and the like) to permit detachment of catalytic converters from vehicles.

[0005] In one aspect, a catalytic converter anti-theft system, according to an example embodiment, for an exhaust system of a vehicle is disclosed. The system includes a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system; a cable configured to be disposed about the thermally insulating first layer; a cut and abrasion resistant second layer configured to be disposed about the cable; and an abrasion resistant third layer configured to be disposed about the cut and abrasion resistant second layer.

[0006] In another aspect, a catalytic converter anti-theft system, according to another example embodiment, for an exhaust system of a vehicle is disclosed. The system includes a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system; a cut and abrasion resistant second layer configured to be disposed about the thermally insulating first layer; an abrasion resistant third layer configured to be disposed about the cut and abrasion resistant second layer; and a plurality of fasteners configured to couple the anti-theft system to the vehicle.

[0007] In yet another aspect, a method for coupling a catalytic converter anti-theft system to an exhaust system of a vehicle, according to yet another example embodiment, is disclosed. The method includes: positioning a thermally insulating first layer about at least a portion of the exhaust system; positioning a cable about the thermally insulating first layer; coupling the cable to the vehicle; positioning a cut and abrasion resistant second layer about the cable; positioning an abrasion resistant third layer about the cut and abrasion resistant second layer; and coupling the abrasion resistant third layer to the vehicle using a plurality of fasteners.

[0008] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS/FIGURES

[0009] Non-limiting and non-exhaustive examples are described with reference to the following

Figures.

[0010] FIG. 1 shows a diagrammatic view of a catalytic converter anti-theft system, according to an example embodiment, coupled to an exemplary exhaust system of an automotive vehicle.

[0011] FIG. 2 shows a detail diagrammatic view of the catalytic converter anti-theft system and the vehicle of FIG. 1.

[0012] FIG. 3 shows a cross-sectional view of the catalytic converter anti-theft system taken along line 3-3 of FIG. 2.

[0013] FIG. 4 shows a cross-sectional view of a cable of the catalytic converter anti-theft system of FIG. 1.

[0014] FIG. 5 shows a cross-sectional view of the catalytic converter anti-theft system taken along line 5-5 of FIG. 2.

[0015] FIG. 6 shows a flow diagram of a method for coupling a catalytic converter anti-theft system to an exhaust system of a vehicle, according to an example embodiment.

[0016] FIG. 7 shows a thermally insulating first layer and fasteners positioned about an exhaust conduit, according to the method of FIG. 6.

[0017] FIG. 8 shows a cut and abrasion resistant second layer and fasteners positioned about the thermally insulating first layer and fasteners of FIG. 7, according to the method of FIG. 6.

[0018] FIG. 9 shows an abrasion resistant third layer and fasteners positioned about the cut and abrasion resistant second layer and fasteners of FIG. 8, according to the method of FIG. 6.

DETAILED DESCRIPTION

[0019] In the following detailed description, references are made to the accompanying drawings that form a part hereof, and in which are shown by way of illustrations specific embodiments or examples. These aspects may be combined, other aspects may be utilized, and structural changes may be made without departing from the present disclosure. Embodiments may be practiced as methods, systems, or devices. The following detailed description is not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims and their equivalents.

[0020] FIG. 1 shows a diagrammatic view of an automotive vehicle **10**, for example a commercial vehicle or a passenger vehicle, and a catalytic converter anti-theft system **100**, according to an example embodiment, coupled to an exhaust system **12** of the vehicle **10**. As used herein, the term “vehicle” and “automotive vehicle” broadly refer to any motor vehicle that includes an exhaust system, including cars, trucks, buses, vans, etc. Illustratively, the exhaust system **12** of the vehicle **10** includes an exhaust manifold **14** in fluid communication with and receiving discharged gases from an engine (not shown) of the vehicle **10**, for example an internal combustion engine such as a diesel engine or a gasoline engine. The illustrative exhaust system **12** further includes a first or upstream exhaust conduit **16** in fluid communication with the exhaust manifold **14**, a catalytic converter **18** in fluid communication with the upstream exhaust conduit **16**, and a second or downstream exhaust conduit **20** in fluid communication with the catalytic converter **18**.

Illustratively, the exhaust system **12** may further include a resonator **22** in fluid communication with the downstream exhaust conduit **20**, a third or “catback” exhaust conduit **24** in fluid communication with the resonator **22**, a muffler **26** in fluid communication with the catback exhaust conduit **24**, and a tailpipe **28** in fluid communication with the muffler **26** for discharging exhaust gases from the vehicle **10**. In some cases, the exhaust system **12** may include additional or alternative components.

[0021] With continued reference to FIG. 1 and additional reference to FIG. 2, the catalytic converter anti-theft system **100** is disposed about at least a portion, or one or more components, of the exhaust system **12**. Illustratively, the catalytic converter anti-theft system **100** includes a first section **102** that is disposed about at least a portion of the first exhaust conduit **16**, a second section **104** that is disposed about the catalytic converter **18**, and a third section **106** that is disposed about at least a portion of the second exhaust conduit **20**. The first section **102**, the second section **104**,

and the third section **106** of the system **100** together deter unauthorized detachment of the catalytic converter **18** from the vehicle **10** by inhibiting cutting through the first exhaust conduit **16** and the second exhaust conduit **20**.

[0022] With further reference to FIG. 2 and additional reference to FIG. 3, the first section **102** of the catalytic converter anti-theft system **100** is as follows. The first section **102** includes a thermally insulating first layer **108** (FIG. 3), or inner layer, that is disposed about the first exhaust conduit **16**. In some embodiments, the thermally insulating first layer **108** is a ceramic fiber fabric or blanket. In certain embodiments, the thermally insulating first layer **108** includes one or more of the following properties: fire-proof at a constant temperature of 2,400 degrees Fahrenheit; flexible to facilitate positioning about the first exhaust conduit **16**; light weight; low heat transfer; sufficient for use in high-temperature applications, such as in kilns, pizza ovens, furnaces, and forges. In certain embodiments, the thermally insulating first layer **108** has a thickness of about 1 inch. In some embodiments, the thermally insulating first layer **108** has an area density of about 542 g per square meter. In some embodiments, the first layer **108** is coupled to the first exhaust conduit **16** via one or more fasteners (shown elsewhere), more specifically one or more non-reversibly releasable fasteners (as used herein, non-reversibly releasable fasteners refer to fasteners that are only releasable by damaging or destroying the fasteners and therefore are not reusable), such as cable ties, more specifically metallic cable ties, and even more specifically stainless steel cable ties.

[0023] The first section **102** of the catalytic converter anti-theft system **100** further includes one or more cables, illustratively a first cable **110** and a second cable **112**, disposed about the thermally insulating first layer **108** and thereby the first exhaust conduit **16**. In some embodiments, the first cable **110** and the second cable **112** extend in a figure-eight configuration about the thermally insulating first layer **108**; stated another way, the first cable **110** extends in a first helical direction about the thermally insulating first layer **108** and the second cable **112** extends in a second, opposite helical direction about the thermally insulating first layer **108**. In certain embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, are coupled to the frame **30** of the vehicle **10** via one or more weldments **114**, such as tack weldments formed in a metal inert gas ("MIG") welding process. In some embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, are metallic cables, more specifically galvanized cables. In some embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, have a diameter of about 3/16 inch. In some embodiments and with additional reference to FIG. 4, one or more of the cables, for example the first cable **110** and/or the second cable **112**, have a 7×19 construction (that is, the cable includes 7 strands **116** each having 19 wires **118**, the cable thereby having a total of 133 wires **118**). In certain embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, include one or more of the following properties: flexible to facilitate positioning about the thermally insulating first layer **108**; corrosion-resistant; cut-resistant. In some embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, are secured or further secured to the exhaust system **12** via one or more fasteners (shown elsewhere), more specifically one or more non-reversibly releasable fasteners, such as cable ties, more specifically metallic cable ties, and even more specifically stainless steel cable ties.

[0024] Referring again to FIG. 3, the first section **102** of the catalytic converter anti-theft system **100** further includes a cut and abrasion resistant second layer **120**, or intermediate layer, that is disposed about the one or more cable(s), for example the first cable **110** and the second cable **112**, and thereby the thermally insulating first layer **108** and the first exhaust conduit **16**. In some embodiments, the cut and abrasion resistant second layer **120** is an aramid fiber layer, more specifically a para-aramid fiber layer, for example a Kevlar® layer, a meta-aramid fiber layer, for example a Nomex® layer, or a layer including both para-aramid fibers and meta-aramid fibers. In some embodiments, the second layer **120** is formed as an interwoven twill weave. In certain embodiments, the cut and abrasion resistant second layer **120** includes one or more of the following

properties: fire-retardant up to 700 degrees Fahrenheit; water repellant; flexible; lightweight; sufficient for use in protective clothing, such as chainsaw pants, gloves, firefighter suits, and body armor. In some embodiments, the cut and abrasion resistant second layer **120** is a 50% Nomex® and a 50% Kevlar® cross weave twill pattern. In certain embodiments, the cut and abrasion resistant second layer **120** has an area density of about 237 g per square meter. In some embodiments, the second layer **120** is secured or further secured to the exhaust system **12** via one or more fasteners (shown elsewhere), more specifically one or more reversibly releasable fasteners (as used herein, reversibly releasable fasteners refer to fasteners that are releasable without damaging or destroying the fasteners, thereby permitting their reuse), such as worm drive clamps, more specifically metallic worm drive clamps, and even more specifically galvanized worm drive clamps.

[0025] With further reference to FIGS. **2** and **3**, the first section **102** of the catalytic converter anti-theft system **100** further includes an abrasion resistant third layer **122** disposed about the cut and abrasion resistant second layer **120**, and thereby the cable(s), for example the first cable **110** and/or the second cable **112**, the thermally insulating first layer **108**, and the first exhaust conduit **16**. In some embodiments, the abrasion resistant third layer **122** is a carbon fiber layer. In some embodiments, the abrasion resistant third layer **122** is formed as an interwoven twill weave, more specifically a 2×2 twill weave. In certain embodiments, the abrasion resistant third layer **122** includes one or more of the following properties: water repellant; flexible; lightweight; high tensile strength; high durability; sufficient for use as a structural reinforcement. In certain embodiments, the abrasion resistant third layer **122** has an area density of about 220 g per square meter. In some embodiments, the abrasion resistant third layer **122** has a thickness of 0.4 mm to 0.5 mm, more specifically about 0.46 mm. In some embodiments, the third layer **122** is secured or further secured to the exhaust system **12** via one or more fasteners **124** (FIG. **2**), more specifically one or more reversibly releasable fasteners, such as worm drive clamps, more specifically metallic worm drive clamps, and even more specifically galvanized worm drive clamps. In certain embodiments, the third layer **122** is coupled to the second layer **120** in alternative or additional manners, for example fusing or stitching together the edges of the layers.

[0026] With reference again to FIG. **2**, the second section **104** of the catalytic converter anti-theft system **100** includes one or more cables, more specifically the first cable **110** and the second cable **112** extending continuously from the first section **102** of the system **100**. In certain embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, are disposed about the catalytic converter **18**. In some embodiments, the first cable **110** and the second cable **112** extend in a figure-eight configuration about the catalytic converter **18**; stated another way, the first cable **110** extends in a first helical direction about the catalytic converter **18** and the second cable **112** extends in a second, opposite helical direction about the catalytic converter **18**. In certain embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, are coupled to the catalytic converter **18** via one or more weldments **126**, such as tack weldments formed in a MIG welding process.

[0027] With reference to FIGS. **2** and **5**, the third section **106** of the catalytic converter anti-theft system **100** is as follows. The third section **106** includes a thermally insulating first layer **128** (FIG. **5**), or inner layer, that is disposed about the second exhaust conduit **20**. In some embodiments, the thermally insulating first layer **128** is the same or substantially similar to the thermally insulating first layer **108** of the first section **102** of the system **110** (that is, having the same or substantially similar features, properties, and/or characteristics as the thermally insulating first layer **108**). In some embodiments, the first layer **128** is coupled to the second exhaust conduit **20** via one or more fasteners (shown elsewhere), more specifically one or more non-reversibly releasable fasteners, such as cable ties, more specifically metallic cable ties, and even more specifically stainless steel cable ties.

[0028] The third section **106** also includes one or more cables, more specifically the first cable **110**

and the second cable **112** extending continuously from the second section **104** of the system **100**. In certain embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, are disposed about the thermally insulating first layer **128** and thereby the second exhaust conduit **20**. In some embodiments, the first cable **110** and the second cable **112** extend in a figure-eight configuration about the thermally insulating first layer **128**; stated another way, the first cable **110** extends in a first helical direction about the thermally insulating first layer **128** and the second cable **112** extends in a second, opposite helical direction about the thermally insulating first layer **128**. In certain embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, are coupled to the frame **30** of the vehicle **10** via one or more weldments **130** (FIG. 2), such as tack weldments formed in a MIG welding process. In some embodiments, one or more of the cables, for example the first cable **110** and/or the second cable **112**, are secured or further secured to the exhaust system **12** via one or more fasteners (shown elsewhere), more specifically one or more non-reversibly releasable fasteners, such as cable ties, more specifically metallic cable ties, and even more specifically stainless steel cable ties.

[0029] The third section **106** of the catalytic converter anti-theft system **100** further includes a cut and abrasion resistant second layer **132** (FIG. 5), or intermediate layer, that is disposed about the one or more cable(s), for example the first cable **110** and the second cable **112**, and thereby the thermally insulating first layer **128** and the second exhaust conduit **20**. In some embodiments, the cut and abrasion resistant second layer **132** is the same or substantially similar to the cut and abrasion resistant second layer **120** of the first section **102** of the system **110** (that is, having the same or substantially similar features, properties, and/or characteristics as the cut and abrasion resistant second layer **120**). In some embodiments, the second layer **132** is secured or further secured to the exhaust system **12** via one or more fasteners (shown elsewhere), more specifically one or more reversibly releasable fasteners (as used herein, reversibly releasable fasteners refer to fasteners that are releasable without damaging or destroying the fasteners, thereby permitting their reuse), such as worm drive clamps, more specifically metallic worm drive clamps, and even more specifically galvanized worm drive clamps.

[0030] The third section **106** of the catalytic converter anti-theft system **100** additionally includes an abrasion resistant third layer **134** disposed about the cut and abrasion resistant second layer **132**, and thereby the cable(s), for example the first cable **110** and/or the second cable **112**, the thermally insulating first layer **128**, and the second exhaust conduit **20**. In some embodiments, the abrasion resistant third layer **134** is the same or substantially similar to the abrasion resistant third layer **122** of the first section **102** of the system **110** (that is, having the same or substantially similar features, properties, and/or characteristics as the abrasion resistant third layer **122**). In some embodiments, the third layer **122** is secured or further secured to the exhaust system **12** via one or more fasteners **136** (FIG. 2), more specifically one or more reversibly releasable fasteners, such as worm drive clamps, more specifically metallic worm drive clamps, and even more specifically galvanized worm drive clamps.

[0031] In certain embodiments, the catalytic converter anti-theft system **100** facilitates maintaining the catalytic converter **18** at an appropriate working temperature. More specifically and as illustrated, one or more of the layers of the first section **102** of the system **100** (for example, the thermally insulating first layer **108**, the cut and abrasion resistant second layer **120**, and/or the abrasion resistant third layer **122**) are disposed apart from, or are not disposed about, the catalytic converter **18**, and one or more of the layers of the third section **106** of the system **100** (for example, the thermally insulating first layer **128**, the cut and abrasion resistant second layer **132**, and/or the abrasion resistant third layer **134**) are disposed apart from, or are not disposed about, the catalytic converter **18**.

[0032] FIG. 6 shows a method **200** for coupling the catalytic converter anti-theft system **100** to an exhaust system of a vehicle, according to an example embodiment, and FIGS. 7-9 show actions associated with the method **200**. The following description of the method **200** describes coupling

components of the first section **102** of the system **100** to the vehicle, although similar actions may be used, simultaneously or non-simultaneously, to couple components of the third section **106** of the system **100** to the vehicle. The method begins at block **202**, and as shown in FIG. 7, by positioning the thermally insulating first layer **108** about at least a portion of the exhaust system, for example the first exhaust conduit **16**. In some embodiments, the thermally insulating first layer **108** is disposed apart from the catalytic converter (shown elsewhere). At block **204** and as shown in FIG. 7, one or more fasteners **138**, such as cable ties, are positioned about the thermally insulating first layer **108** and thereby the first exhaust conduit **16**. At block **206**, one or more cables, for example the first cable **110** and the second cable **112** (shown elsewhere), are positioned about the fasteners **138**, and thereby the thermally insulating first layer **108** and the first exhaust conduit **16**. In certain embodiments, the cables are also positioned about the catalytic converter. In some embodiments, the cables are positioned in a figure-eight configuration about the first exhaust conduit **16** and/or the catalytic converter. At block **208**, the cables, for example the first cable **110** and the second cable **112**, are coupled to the vehicle (shown elsewhere) and/or the catalytic converter, more specifically the cables are welded to the frame of the vehicle and/or the catalytic converter. At block **210** and as shown in FIG. 8, the method **200** continues by positioning the cut and abrasion resistant second layer **120** about the cables and thereby the fasteners **138**, the thermally insulating first layer **108**, and the first exhaust conduit **16**. In some embodiments, the cut and abrasion resistant second layer **120** is disposed apart from the catalytic converter. At block **212** and as shown in FIG. 8, one or more fasteners **140**, such as worm drive clamps, are positioned about the cut and abrasion resistant second layer **120** and thereby the cables, the fasteners **138**, the thermally insulating first layer **108**, and the first exhaust conduit **16**. At block **214** and as shown in FIG. 9, the method **200** continues by positioning the abrasion resistant third layer **122** about the fasteners **140** and thereby the cut and abrasion resistant second layer **120**, the cables, the fasteners **138**, the thermally insulating first layer **108**, and the first exhaust conduit **16**. In some embodiments, the abrasion resistant third layer **122** is disposed apart from the catalytic converter. At block **216** and as shown in FIG. 9, one or more of the fasteners **124** are positioned about the abrasion resistant third layer **122** and thereby the fasteners **140**, the cut and abrasion resistant second layer **120**, the cables, the fasteners **138**, the thermally insulating first layer **108**, and the first exhaust conduit **16**.

[0033] The description and illustration of one or more aspects provided in this application are not intended to limit or restrict the scope of the following claims in any way. The aspects, examples, and details provided in this application are considered sufficient to convey possession and enable others to make and use embodiments according to the present disclosure. The following claims should not be construed as being limited to any aspect, example, or detail provided in this application. Regardless of whether shown and described in combination or separately, the various features (both structural and methodological) are intended to be selectively included or omitted to produce an embodiment with a particular set of features. Having been provided with the description and illustration of exemplary embodiments, one skilled in the art may envision variations, modifications, and alternate aspects falling within the spirit of the broader aspects of the disclosure of this application that do not depart from the scope of the following claims.

[0034] The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the disclosed embodiments and does not pose a limitation on the scope of the disclosed embodiments unless otherwise claimed. Numerous modifications and adaptations will be readily apparent to those of ordinary skill in this art.

Claims

1. A catalytic converter anti-theft system for an exhaust system of a vehicle, the system comprising: a thermally insulating first layer configured to be disposed about at least a portion of the exhaust

- system; a cable configured to be disposed about the thermally insulating first layer; a cut and abrasion resistant second layer configured to be disposed about the cable; and an abrasion resistant third layer configured to be disposed about the cut and abrasion resistant second layer.
2. The anti-theft system of claim 1, wherein the thermally insulating first layer is a ceramic fiber blanket.
 3. The anti-theft system of claim 1, wherein the cable is a galvanized cable.
 4. The anti-theft system of claim 1, wherein the cut and abrasion resistant second layer is an aramid fiber layer.
 5. The anti-theft system of claim 1, wherein the abrasion resistant third layer is a carbon fiber layer.
 6. The anti-theft system of claim 1, further comprising a plurality of fasteners configured to couple the anti-theft system to the vehicle.
 7. The anti-theft system of claim 6, wherein the plurality of fasteners comprise worm drive clamps disposed about the abrasion resistant third layer.
 8. The anti-theft system of claim 6, wherein the plurality of fasteners is a first plurality of fasteners, the first plurality of fasteners being disposed about the abrasion resistant third layer, and further comprising a second plurality of fasteners, the second plurality of fasteners being disposed about the thermally insulating first layer.
 9. The anti-theft system of claim 8, wherein the second plurality of fasteners comprise cable ties.
 10. The anti-theft system of claim 1, wherein the thermally insulating first layer is a first thermally insulating first layer, the cut and abrasion resistant second layer is a first cut and abrasion resistant second layer, and the abrasion resistant third layer is a first abrasion resistant third layer, and wherein the anti-theft system comprises: a first section comprising the first thermally insulating first layer, the first cut and abrasion resistant second layer, the cable, and the first abrasion resistant third layer; a second section comprising the cable, the cable extending continuously from the first section and configured to be disposed about at least a portion of the exhaust system; a third section comprising: a second thermally insulating first layer configured to be disposed about at least a portion of the exhaust system; the cable, the cable extending continuously from the second section and configured to be disposed about the second thermally insulating first layer; a second cut and abrasion resistant second layer configured to be disposed about the cable; and a second abrasion resistant third layer configured to be disposed about the second cut and abrasion resistant second layer.
 11. The anti-theft system of claim 10, wherein the cable is a first cable, and further comprising a second cable configured to be disposed about the first thermally insulating first layer in the first section, the second cable extending continuously from the first section to the second section and configured to be disposed about at least a portion of the exhaust system in the second section, the second cable extending continuously from the second section to the third section and configured to be disposed about the second thermally insulating first layer in the third section.
 12. A catalytic converter anti-theft system for an exhaust system of a vehicle, the system comprising: a thermally insulating first layer configured to be disposed about at least a portion of the exhaust system; a cut and abrasion resistant second layer configured to be disposed about the thermally insulating first layer; an abrasion resistant third layer configured to be disposed about the cut and abrasion resistant second layer; and a plurality of fasteners configured to couple the anti-theft system to the vehicle.
 13. The anti-theft system of claim 12, wherein the thermally insulating first layer is a ceramic fiber blanket.
 14. The anti-theft system of claim 12, wherein the cut and abrasion resistant second layer is an aramid fiber layer.
 15. The anti-theft system of claim 12, wherein the abrasion resistant third layer is a carbon fiber layer.
 16. The anti-theft system of claim 12, wherein the plurality of fasteners comprise worm drive

clamps disposed about the abrasion resistant third layer.

17. The anti-theft system of claim 12, wherein the plurality of fasteners is a first plurality of fasteners, the first plurality of fasteners being disposed about the abrasion resistant third layer, and further comprising a second plurality of fasteners, the second plurality of fasteners being disposed about the thermally insulating first layer.

18. The anti-theft system of claim 17, wherein the second plurality of fasteners comprise cable ties.

19. A method for coupling a catalytic converter anti-theft system to an exhaust system of a vehicle, the method comprising: positioning a thermally insulating first layer about at least a portion of the exhaust system; positioning a cable about the thermally insulating first layer; coupling the cable to the vehicle; positioning a cut and abrasion resistant second layer about the cable; positioning an abrasion resistant third layer about the cut and abrasion resistant second layer; and coupling the abrasion resistant third layer to the vehicle using a plurality of fasteners.

20. The method of claim 19, wherein the exhaust system comprises an exhaust conduit and a catalytic converter coupled to the exhaust conduit, and the thermally insulating first layer, the cut and abrasion resistant second layer, and the abrasion resistant third layer are positioned about the exhaust conduit and disposed apart from the catalytic converter.

21. The method of claim 20, wherein coupling the cable to the vehicle comprises welding the cable to the catalytic converter.

22. The method of claim 20, wherein coupling the cable to the vehicle comprises welding the cable to a frame of the vehicle.

23. The method of claim 19, wherein the cable is a first cable, and positioning the first cable about the thermally insulating first layer comprises positioning the first cable and a second cable about the thermally insulating first layer and a catalytic converter of the exhaust system.

24. The method of claim 23, wherein positioning the first cable and the second cable about the thermally insulating first layer and the catalytic converter comprises positioning the first cable and the second cable in a figure-eight configuration about the thermally insulating first layer and the catalytic converter.

25. The method of claim 19, wherein the plurality of fasteners comprise worm drive clamps, and coupling the abrasion resistant third layer to the vehicle comprises positioning the worm drive clamps about the abrasion resistant third layer.
